ALICE L WALTON SCHOOL OF MEDICINE

Megaloblastic Anemia

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Adapted from: Megaloblastic Anemia

Listen to this Brick

0:00 / 27:23 1x

Learning Objectives (4) | Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Describe the pathophysiology of the mentioned anemia.
3. Interpret clinical findings to properly diagnose the mentioned anemia.
4. Outline the [illegible]

CASE CONNECTION

TD is a 50-year-old female referred from the hematology clinic with fatigue, numbness, and tingling of the lower extremities. On physical examination, TD has decreased sensation to pinprick and vibration of the legs. Her hemoglobin is low and her mean corpuscular volume (MCV) is high. Peripheral blood smear shows a neutrophil with a six-lobed nucleus and abnormally large red blood cells.

What other blood tests should be ordered to make the definitive diagnosis? How will you explain the sensory symptoms to TD? Consider your answers as you read, and we'll revisit TD at the end of the brick.

GO TO CONCLUSION ↓

What Is Megaloblastic Anemia?

Megaloblastic anemia is a type of anemia caused by vitamin B_{12} or folate (vitamin B_9) deficiency. It is characterized by enlarged **macrocytic erythrocytes** in the peripheral blood (macrocytic anemia), along with multilobed white blood cells in the peripheral smear and a bone marrow showing large erythroid progenitor cells with immature nuclear chromatin.

The anemia is caused by lack of vitamin cofactors for DNA synthesis, leading to poor bone marrow red blood cell (RBC) production—ineffective hematopoiesis.

Vitamin B_{12} and folate deficiencies are the two most common causes of megaloblastic anemi[illegible]ibit DNA synthesis and cause this anemia. Therefore, this type of anemia is more common in patients older than age 40 years, since the prevalence of these vitamin deficiencies increases with age. Together, vitamin B_{12} and folate deficiency accounts for about 2% of all anemia cases. New cases of megaloblastic anemia per year are decreasing because of vitamin supplementation in pregnant and older patients. These deficiencies are rare in children.

How Do Patients with Megaloblastic Anemia Present (LO #1)?

Most patients with mild anemia are usually asymptomatic. However, patients with any type of moderate-to-severe anemia may experience shortness of breath, fatigue, weakness, dizziness/lightheadedness, and tachycardia.

Some patients may report specific symptoms related to the underlying cause of the anemia. For example, patients with vitamin B_{12} deficiency may report neurologic symptoms such as numbness and tingling in their hands and feet.

CLINICAL CORRELATION

Folate deficiency is usually not associated with neurologic symptoms.

What symptoms will a patient with vitamin B_{12} deficiency likely experience?

CLINICAL CORRELATION

There is a poor correlation between neurologic and hematologic changes in vitamin B12 deficiency. Neurologic manifestations may be present without macrocytosis or anemia; conversely, patients with severe hematologic changes may have no neurologic signs.

The absence of anemia or macrocytosis does not exclude neurologic disease due to vitamin B12 deficiency.

What Is the Pathophysiology of Megaloblastic Anemia (LO #2)?

The mechanism of megaloblastic anemia is **impaired DNA synthesis**, usually related to the deficiency of vitamin B_{12} (cobalamin), folate, or both.

Both vitamin B_{12} and [illegible] is in cells, which is crucial in cells with rapid turnover rate, such as hematopoietic cells and the cells that line the gut. A deficiency of these vitamins will therefore disrupt DNA synthesis and **impede cell division**. However, the cell cytoplasm continues to grow because of normal RNA and protein (hemoglobin) synthesis. This nuclear-cytoplasmic asynchrony leads to **ineffective erythropoiesis** of the developing blood cells in the bone marrow, causing anemia.

Etiology

Shared causes of vitamin B_{12} and folate deficiency include gastrointestinal (GI) **malabsorption syndromes** like celiac and Crohn diseases or surgical bowel resection, which can impair intestinal vitamin absorption. Patients with **heavy alcohol intake** can be deficient in both vitamins, often because of otherwise limited food intake.

CLINICAL CORRELATION

Heavy alcohol use can also cause anemia and macrocytosis without vitamin B_{12} and folate deficiency, so you should always check the vitamin levels in such patients.

In addition, certain conditions can lead to vitamin B_{12} or folate deficiency.

Vitamin B_{12} Deficiency. Specific causes of vitamin B_{12} deficiency include:

- **Pernicious anem**[illegible]rietal cells or intrinsic factor [illegible] of parietal cells leads to decreased gastric acidity and lack of intrinsic factor needed for intestinal vitamin B_{12} absorption. This is usually seen in patients with chronic gastritis. They may be asymptomatic or have indigestion or burning in the upper abdomen.
- **Pancreatic insufficiency**: the pancreatic enzymes needed to free vitamin B_{12} from its transcobalamin "protector" for intestinal absorption are deficient.
- **Drugs**: metformin, H2 blockers, and proton pump inhibitors interfere with vitamin B_{12} absorption.
- **Infection with parasitic worms**: the fish tapeworm *Diphyllobothrium latum* colonizes the terminal ileum and competes for vitamin B_{12} absorption.

Any of the above can be worsened by a diet that is lacking in common sources of vitamin B_{12} such as animal liver, meat, fish, cheese, dairy, eggs, and so on (this includes strict vegan and some fad diets).

Folate Deficiency. Specific causes of folate deficiency include:

- Gastric bypass surgery
- A diet lacking in common sources of folate, including dark green leafy vegetables such as lettuce, brussels sprouts, or spinach
- Medications such as methotrexate and trimethoprim (antimetabolites) that inhibit the conversion of folate to its active form and anti-epileptic drugs like phenytoin that inhibit folate absorption in the duodenum

The liver has a large store of vitamin B_{12}, which typically takes years of poor intake or absorption to be depleted. On the other hand, folate deficiency may present with only 4 to 5 months of a folate-depleted diet. Because body folate reserves are not as su[illegible]ditions that increase cell proliferation and require rapid utilization of folate for DNA synthesis can also cause folate deficiency (even when folate intake is sufficient). These include:

- Pregnancy (folate is in prenatal vitamins)
- Chronic hemolytic anemias (in which RBC synthesis must increase to meet the demand of RBC loss)
- Cancer (rapid cell turnover)

CLINICAL CORRELATION

Folate deficiency in women in early pregnancy is associated with a higher risk of neural tube defects in the baby. In order to decrease this occurrence, the FDA mandated that all grain products in the US be supplemented with folate. Additionally, it is very important for women who are or are planning to become pregnant take prenatal vitamins (rich in folate).

Medications

Apart from vitamin deficiencies, several drugs are known to inhibit DNA synthesis and cause megaloblastic anemia. These most commonly include chemotherapy and antiretroviral drugs and hydroxyurea, used for sickle cell disease.

Table 1

	[illegible]	Folate
Food Source	Animal products	Grains, fruits, and vegetables
Main Cause of Deficiency	Malabsorption (dietary deficiency rare)	Dietary deficiency and ethanol (malabsorption rare)

Pathogenesis

Vitamin B_{12} and folate are critical in working together to allow DNA synthesis. Decreased DNA synthesis due to vitamin deficiencies may cause lack of cell proliferation, especially with rapidly produced cells like RBCs. Mechanisms include:

- Vitamin B_{12} is a cofactor for conversion of homocysteine to methionine by methionine synthase. Methionine is required for DNA synthesis. In so doing, it donates its methyl group and becomes inactive.
- Folate in its active form (methyl tetrahydrofolate [THF]) donates its methyl group to vitamin B_{12}, reactivating it.
- Folate is also needed in the form of methylene-THF to synthesize thymidine, one of the building blocks of DNA. Folate is also needed to synthesize the purines adenosine and guanosine.
- In addition, if vitamin B_{12} is deficient, folate is "trapped" in the form of methyl-THF. This causes the levels of THF and methylene-THF to drop, and DNA synthesis is impaired.

While both vitamins are critical for RBC production, only vitamin B_{12} deficiency produces neurologic deficits such as tingling and confusion. This is because vitamin B_{12} (but not folate) is required for the conversion of methylmalonyl CoA to succinyl CoA by methylmalonyl-CoA mutase, a step toward **myelin synth**[illegible]on (Figure 1).

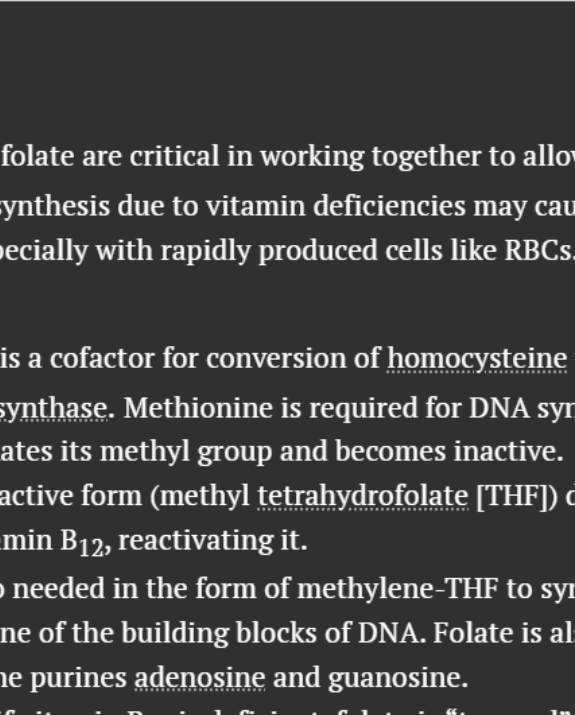


Figure 1

Why does a deficiency of vitamin B_{12}, but not folate, commonly lead to

Pathology

In megaloblastic anemia, the peripheral blood smear shows **enlarged RBCs** (macrocytes). These can also be seen in other anemias (e.g., due to heavy alcohol use).

More specific to megaloblastic anemia is the presence of enlarged RBCs with an oval shape (macro-ovalocytes), seen in Figure 2.

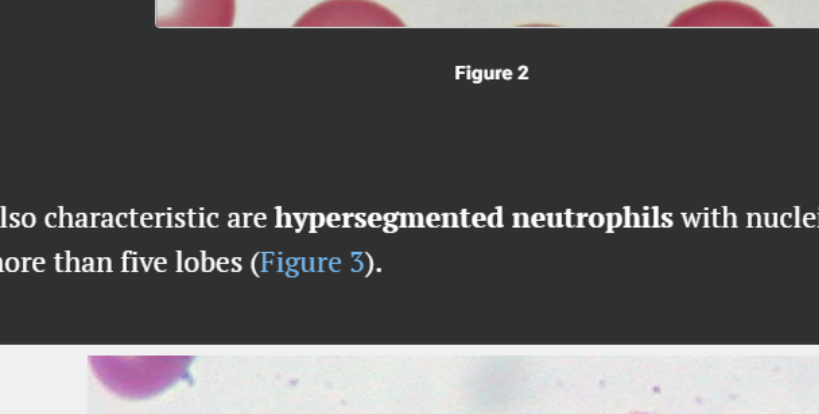
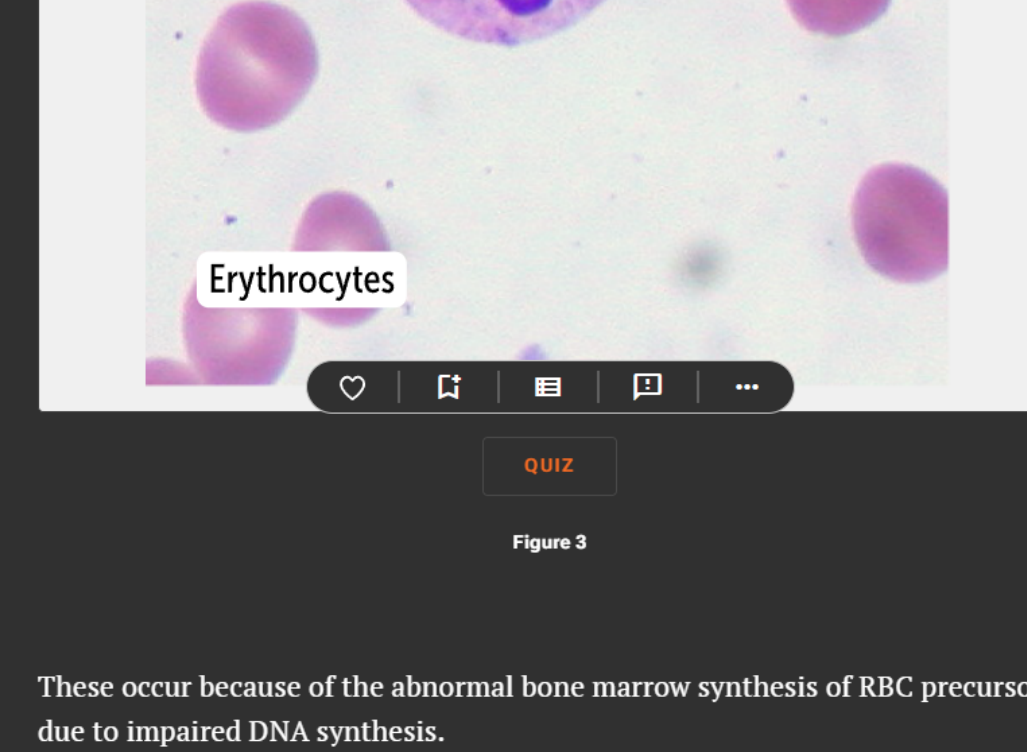


Figure 2

Also characteristic are **hypersegmented neutrophils** with nuclei containing more than five lobes (Figure 3).



QUIZ

Figure 3

These occur because of the abnormal bone marrow synthesis of RBC precursors due to impaired DNA synthesis.

While bone marrow biopsy is not routinely done to diagnose megaloblastic anemia, the bone marrow is hypercellular in appearance, with increased numbers of enlarged erythroid and myeloid precursors and megakaryocytes. Dysplastic erythroblasts with abnormally shaped nuclei can also be seen.

What are bone marrow findings in patients with megaloblastic anemia?

How Do We Dia[illegible]nia (LO #3)?

As with other forms of anemias, the diagnosis of megaloblastic anemia is based on a combination of clinical history, physical exam findings, and lab tests. It should be suspected in anemic patients with any of the above conditions (e.g., dietary limitations, GI disease, cancer).

Physical Examination

A physical exam feature specific to **vitamin B_{12} deficiency** is the presence of neurologic deficits. The most common neurologic problems are numbness and gait disturbances. **Subacute combined degeneration** of the spinal cord can also be seen, in which the spinal cord, dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts are damaged, leading to sensory deficits, ataxia, and hyperreflexia. A variety of other cognitive, mood, and visual disturbances have also been described.

Folate deficiency can cause **oral ulcers**, due to the rapid division of the oral mucosa, which drops with decreased DNA synthesis. Cognitive or mood deficits can also be seen. Pregnant women without sufficient folate supplementation are more likely to give birth to infants with low birth weight and with neural tube defects such as spina bifida. Glossitis (tongue pain, swelling, tenderness) can be seen in both vitamin B12 and folate deficiencies.

Laboratory Tests

The diagnosis is usually made with characteristic findings on the complete blood count (CBC) and blood smear, in the right clinical setting.

Complete Blood Cou[illegible]**macrocytic**, with a high mean corpuscular [illegible]

CLINICAL CORRELATION

A high mean corpuscular volume is not diagnostic alone, since macrocytic anemia can occur without megaloblastic anemia. Such causes include heavy alcohol use, micronutrient deficiencies (e.g., copper), hypothyroidism, and certain malignancies (i.e. myelodysplastic neoplasms).

CLINICAL CORRELATION

An additional diagnosis on the differential of macrocytic anemia is **Diamond-Blackfan anemia (DBA).** Presenting in infancy, this disorder is a congenital erythroid aplasia characterized by a progressive macrocytic anemia, reticulocytopenia, and cancer predisposition (leukemia, myelodysplastic neoplasms, osteosarcoma). Importantly, it is associated with other congenital abnormalities (craniofacial abnormalities, triphalyngeal thumbs, growth failure, and/or developmental delays). In contrast to aplastic anemia (decrease in all cell lines, leading to pancytopenia), DBA causes a pure red cell aplasia (only affects the red cell line).

Severe macrocytosis (MCV >115 fL) is commonly seen only in megaloblastic anemia.

The reticulocyte coun[illegible]not being made normally (ineffective [illegible]

Other CBC findings can include low white blood cell or platelet counts due to poor bone marrow precursor production.

Peripheral Blood Smear. Characteristic features are **macro-ovalocytes** (see Figure 2) and **hypersegmented neutrophils** (see Figure 3). These are not seen in other causes of macrocytic anemia like heavy alcohol use.

Serum Studies. Once the diagnosis of megaloblastic anemia is made, the next step in determining the cause is measuring **serum vitamin B_{12} and folate levels.** A low level of either or both makes the diagnosis. Patients with megaloblastic anemia may also have laboratory studies suggestive of hemolysis (increased LDH, unconjugated bilirubin; decreased haptoglobin).

Pregnancy, multiple myeloma, and HIV infection can falsely lower vitamin B_{12} levels, while renal or liver disease and some malignancies can falsely raise them.

Because of concerns about accuracy, low vitamin B_{12} levels are often confirmed using serum levels of methylmalonic acid (MMA) and homocysteine. These should both be high in vitamin B_{12} deficiency because of decreased cellular metabolism.

Serum anti-intrinsic factor and antiparietal cell antibodies should be ordered in patients with vitamin B_{12} deficiency who may have gastritis and pernicious anemia or who have vitamin B_{12} deficiency without an obvious cause.

What are the typical findings of megaloblastic anemia on the complete blood count (CBC) and blood smear?

How Do We Manage Megaloblastic Anemia (LO #4)?

Most cases of megaloblastic anemia respond to appropriate vitamin supplementation or changes in the diet if it is limited in vitamin B_{12} or folate. Those with alcohol use disorder should stop consuming alcohol.

For folate deficiency, treatment is **folic acid replacement** until the hematologic abnormalities seen on CBC resolve. Note that folic acid supplementation alone may reverse the hematologic abnormalities of vitamin B_{12} deficiency, but it will often not reverse the neurologic symptoms.

For vitamin B_{12} deficiency, replacement can be with either oral or intramuscular **vitamin B_{12}**. If deficiency is due to impaired GI absorption (e.g., pernicious anemia) the intramuscular route is required, and patients may require supplemental vitamin B_{12} for life.

Why should patients with folate deficiency also be tested for vitamin B_{12} deficiency?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TD, what blood tests should be ordered? How do you explain the sensory symptoms to her?

TD has the classic findings of megaloblastic anemia: macrocytosis and neutrophils with hypersegmented nuclei. Blood levels are ordered; vitamin B_{12} is low and folate is normal. You explain this to TD, who asks, "I can see why being anemic makes me tired, but why do my legs feel weird?" You explain that vitamin B_{12} is necessary for production of the coating on nerve fibers; this production is impaired if vitamin B_{12} is deficient, causing nerves to degenerate. You tell TD, "The good news is that, with treatment, this will improve." When TD is seen in follow-up in 6 weeks, her hemoglobin is in the normal range, and her numbness and tingling have resolved.

Summary

- Megaloblastic anemia is a type of anemia caused by vitamin B12 or folate deficiency, leading to impaired DNA synthesis and ineffective hematopoiesis.
- Folate is needed in the form of methylene-THF to synthesize thymidine, one of the building blocks of DNA. Vitamin B12 is required to regenerate this form of folate for this reaction.
- Vitamin B12 is a cofactor for the conversion of homocysteine to methionine by methionine synthase. Methionine is required for DNA synthesis. It is through this reaction that methylene-THF will ultimately get regenerated.
- Common symptoms include shortness of breath, fatigue, weakness, and in cases of vitamin B12 deficiency, neurologic symptoms such as numbness and tingling in extremities.
- Megaloblastic anemia is more common in patients older than 40 years due to increased prevalence of vitamin deficiencies with age.
- The diagnosis is based on clinical history, physical exam findings, and lab tests, with characteristic findings on the complete blood count (CBC) and blood smear.
- Specific causes of vitamin B12 and folate deficiency include dietary limitations, gastrointestinal disease, cancer, and certain medications.
- Treatment involves addressing the underlying cause and replacing the deficient vitamins.

Review Questic[illegible]

1. A 75-year-old male comes to the physician because of a 2-month history of diffuse weakness and tingling of his arms and legs. Neurologic examination shows weakness of all muscles of the lower extremities. Knee and ankle deep-tendon reflexes are increased. Vibration and position senses are decreased in all extremities. His complete blood count is significant for a hemoglobin of 9.4 g/dL and a mean corpuscular volume of 107 fL (normal 80-100 fL). Which of the following is most likely elevated in this patient?

A. ○ Methionine
B. ● Methylmalonic acid
C. ○ Succinyl CoA
D. ○ Tetrahydrofolate
E. ○ Vitamin B_{12}

Hide Explanation

The correct answer is methylmalonic acid (B). This patient has classic symptoms and signs of subacute combined degeneration with paresthesia, weakness, hyperreflexia, and decreased vibration and proprioception. Vitamin B_{12} deficiency is associated with subacute combined degeneration because it is a cofactor in metabolic reactions crucial to myelin synthesis. Vitamin B_{12} is also required for the conversion of methyl[illegible]Therefore, the impediment of this metabolic step in vitamin B_{12} deficiency will lead to an accumulation of methylmalonyl CoA and, subsequently, elevated methylmalonic acid. The levels of methionine (A), succinyl CoA (C), tetrahydrofolate (D), and vitamin B_{12} (E) are low in vitamin B_{12} deficiency.

2. Which of the following causes of megaloblastic anemia will respond best to oral folate supplements?

A. ○ Chronic pancreatitis
B. ○ Crohn disease
C. ○ Gastritis
D. ○ Pernicious anemia
E. ● Pregnancy

Hide Explanation

The correct answer is pregnancy (E). Pregnancy causes high cell turnover and requires high DNA replication rates, so patients can become folate deficient, especially if their diet is inadequate. Oral folate will correct the problem (as part of good prenatal care). Chronic pancreatitis (A), atrophic gastritis (C), and pernicious anemia (D) are more likely to lead to vitamin B_{12} deficiency, not folate deficiency. Crohn disease (B) is a malabsorptive disease for which oral vitamin supplements would not be effectiv[illegible]e used instead.

3. A 25-year-old female presents to the clinic because of 1 year of worsening fatigue, weakness, and difficulty maintaining her balance. She has a history of alcohol use disorder. When she went to a clinic 2 weeks ago, she was started on folate supplementation. At that time, her physical exam showed a smooth, glossy tongue and decreased vibration sense bilaterally, with an unsteady gait. Her complete blood count showed a hemoglobin of 9 g/dL and a mean corpuscular volume of 106 fL. Which of the following findings will be most likely be seen on today's exam?

A. ● Improvement of anemia but unchanged neurologic symptoms
B. ○ Improvement of both anemia and neurologic symptoms
C. ○ Symptoms remain unchanged
D. ○ Worsening anemia but improvement of neurologic symptoms
E. ○ Worsening of both anemia and neurologic symptoms

Hide Explanation

The correct answer is improvement of anemia but unchanged neurologic symptoms (A). This patient has symptoms of vitamin B_{12} deficiency, likely from her history of alcohol use disorder, including the glossitis, neurologic findings, and macrocytic anemia. Giving folate [illegible]e neurologic symptoms, which are [illegible]ncy. Improvement of both anemia and neurologic symptoms (B) is incorrect because this outcome would be expected if she was given vitamin B_{12}. The patient's anemia will respond to folate, which means her symptoms remain unchanged (C) is incorrect. With folate supplementation, she should not experience worsening anemia but improvement of neurologic symptoms (D) nor worsening of both anemia and neurologic symptoms (E).

4. Which findings on peripheral smear would be suggestive of megaloblastic anemia?

A. ○ Increased schistocytes (red blood cell fragments)
B. ○ Microcytic, hypochromic red blood cells with increased elliptocytes
C. ● Macro-ovalocytes and hypersegmented neutrophils
D. ○ Spherocytes and increased reticulocytes
E. ○ Teardrop cells

Hide Explanation

The correct answer is macro-ovalocytes and hypersegmented neutrophils (C). Increased schistocytes (A) is seen in microangiopathic hemolytic anemia. Microcytic, hypochromic red blood cells with increased elliptocytes (B) is seen in iron deficiency anemia. Spherocytes and increased reticulocytes (D) is seen in hereditary spherocyt[illegible]ardrop cells (E) are seen in myelofibrosis as well as any other space-occupying lesion in the bone marrow.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of the mentioned anemia.~~
02 ☑ ~~Describe the pathophysiology of the mentioned anemia.~~
03 ☑ ~~Interpret clinical findings to properly diagnose the mentioned anemia.~~
04 ☑ ~~Outline the management of the mentioned anemia.~~

Objective Confidence: 4/4